TRANSFORMATIONS · IMAGE AND KERNEL

Image and kernel, side by side

Two subspaces that live in different spaces and answer opposite questions. Every line below states the same attribute for both, and reading them in pairs is what makes the symmetry — and where it breaks — visible.

7
ATTRIBUTES

WHAT EACH ONE IS — 3

1 — Definition

`outputs produced vs inputs destroyed`

$\text{im}(T) = \{T(\mathbf{v})\}$ vs $\ker(T) = \{\mathbf{v} : T(\mathbf{v}) = \mathbf{0}\}$

The image collects what comes out; the kernel collects what goes in and vanishes. Both are subspaces rather than arbitrary sets — which is what makes dimension the right way to measure them.

2 — Where it lives

`codomain vs domain`

$\text{im}(T) \subseteq W$, $\ker(T) \subseteq V$

They sit in different spaces, so they cannot be compared directly, intersected, or added. This is the asymmetry the side-by-side layout most easily hides — the two columns are not two views of one object.

3 — Matrix counterpart

`for` $T(\mathbf{x}) = A\mathbf{x}$

$\text{Col}(A)$ vs $\text{Null}(A)$

The column space and null space under different names. Everything on this page is therefore computable by row reduction, and the abstract statement and the matrix statement never diverge.

HOW EACH IS COMPUTED AND MEASURED — 2

4 — Dimension

`summing to` $n = \dim V$

$\text{rank}(A)$ vs $n - \text{rank}(A)$

Rank-nullity. The two dimensions are not independently chosen — fixing one fixes the other, because every dimension of the domain either survives or collapses. Note the sum is $\dim V$, not $\dim W$: the theorem is about the domain.

5 — How to compute it

`both from one reduction`

pivot columns of the **original** A vs RREF parametric solution

The image basis is taken from A itself — row operations change the column space — while the kernel basis comes from the reduced form. Opposite rules from the same computation, and the pair most often confused.

WHAT THE EXTREME CASES MEAN — 2

6 — Trivial

T `is the zero map vs` T `is injective`

$\text{im}(T) = \{\mathbf{0}\}$ vs $\ker(T) = \{\mathbf{0}\}$

Here the symmetry breaks hardest. A trivial image means the map does nothing at all; a trivial kernel means it loses nothing — the strongest possible statement in the other direction. Same word, opposite significance.

7 — Full

T `is surjective vs` T `is the zero map`

$\text{im}(T) = W$ vs $\ker(T) = V$

And again reversed. A full image is surjectivity, the ideal end; a full kernel is the zero map, the degenerate end. Read with the row above, each column runs from degenerate to ideal, but in opposite directions.

The pairing is not decorative: rank-nullity ties the two dimensions together, so the subspaces are not independent objects that happen to be worth comparing. Every direction of the domain is accounted for exactly once — either it survives into the image or it collapses into the kernel.